Exact polynomial norm and matrix witness equivalence

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Construction. For a field K, finite index set I, supplied polynomial families $A_i, B_i \in K[X]$, and $f \in K[X]$, put

$$A_w = \sum_i w_i A_i, \quad B_w = \sum_i w_i B_i, \quad L(M) = \sum_{i,j} M_{ij}(A_i A_j - f B_i B_j).$$

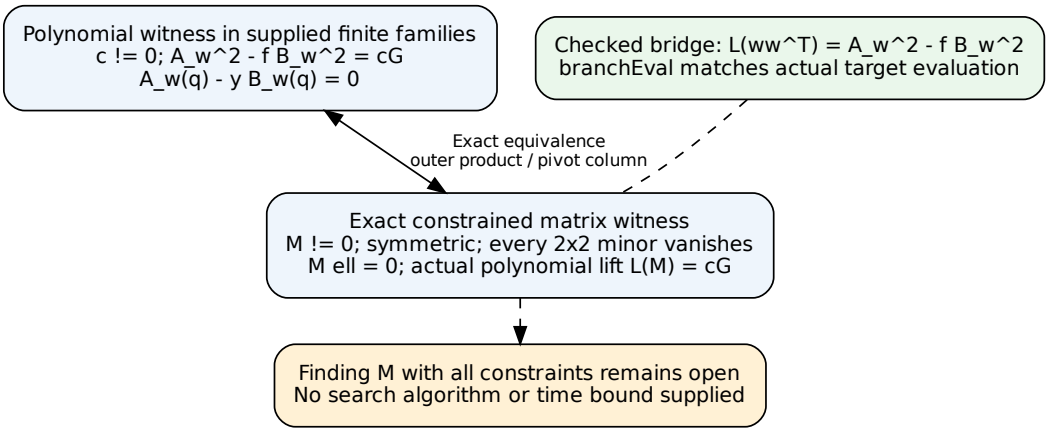
For target coordinates $q, y \in K$, set $\ell_i = A_i(q) - y B_i(q)$. Lean checks the actual polynomial identities

$$L(ww^T) = A_w^2 - f B_w^2, \qquad \sum_i w_i \ell_i = A_w(q) - y B_w(q).$$

Exact witness theorem. If $G \in K[X]$ is nonzero, then the existence of w and $c \neq 0$ with

$$A_w^2 - f B_w^2 = cG, \qquad A_w(q) - y B_w(q) = 0$$

is equivalent to the existence of a nonzero symmetric matrix M and $c \neq 0$ with all two-by-two minors zero, $M\ell = 0$, and $L(M) = cG$. The coefficient vector w is not assumed nonzero: the nonzero norm equation proves that condition. The theorem concerns the spans of the supplied families.



Proof, formal scope and evidence

Forward direction. Use $M = ww^T$. Nonzero cG excludes $w = 0$, so one diagonal entry w_k^2 is nonzero. Symmetry and every minor equation follow from the outer product. The branch identity gives $M\ell = 0$; the norm expansion gives the actual polynomial lift equation.

Reverse direction. For a nonzero symmetric minor-vanishing matrix, a nonzero diagonal pivot M_{kk} exists. Set $w_i = M_{ik}$ and $s = M_{kk}^{-1}$. Then $M = sww^T$, $s \neq 0$. Its pivot row and symmetry give $\sum_i w_i \ell_i = 0$. Linearity of the defined lift gives $A_w^2 - fB_w^2 = (c/s)G$, with $c/s \neq 0$. No square root or extension field is needed. The reverse implication itself does not require $G \neq 0$.

Proof map. Module `NormLift`: `outerNorm`, `branchEval`, `witnessVectorNonzero`, `normToMatrix`, `matrixToNorm`, `witnessEquivalence`. The extraction module additionally supplies `outerMinors`; its earlier five claimed theorem declaration hashes are unchanged. Explicit minor equations are used, without assuming a numerical-rank oracle.

Evidence. Source commit `9cba2446b`; Lean 4.35.0 and pinned manifest. Default build004 passed3143 jobs. Actual tracked-source audit: 28 declarations, zero additional trusted axioms. Printed foundations are `propext`, `Classical.choice`, `Quot.sound` only. Fresh import-inclusive status is recorded in the norm-lift evidence directory's `fresh-status.json`. Builds002/003 failed at an imported definition's module interface and are retained as failures. The public minor constructor resolves that boundary.

Native record `IDC1h2e7ff8ae9d53fc24` supplements the four-component expansion with32 statement-derived and32 extra checker-stream points and a source hash. It does not execute Lean or certify arbitrary arity by sampling. The Lean proof provides the universal finite-index identity. New PCC claim bindings are unsigned and retained in `norm-lift-evidence/claim-pins.json`.

Remaining dependencies. Monomial-basis degree completeness, the complete curve/divisor/group bridge, and an algorithm for finding the constrained matrix are separate. No search-time bound is established. No experiment, performance measurement, target solve or universal lower bound is reported.

Sources. Pinned Mathlib polynomial evaluation and linear-map lemmas; Galbraith's norm/divisor sections linked in `sources.json`; the Nov13 non-Semaev note as scoped context. KB unavailable; no novelty is claimed for this elementary lifting construction. Existing historical graphs remain unchanged.